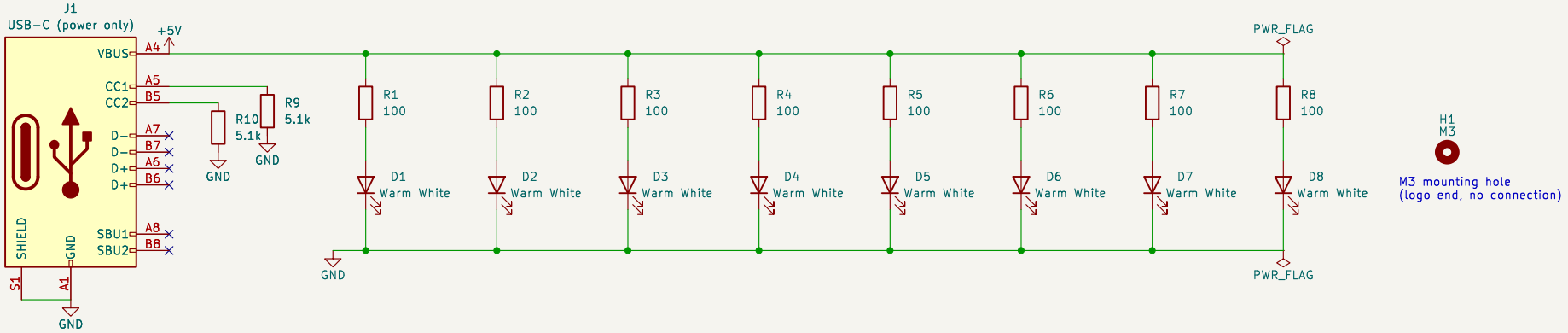
8 LEDs, each with its own resistor (parallel branches).
I(LED) ≈ (5.0 V − 2.9 V) / 100 Ω ≈ 21 mA (typical LED-strip drive current)
Total ≈ 168 mA from USB. P(R) ≈ 44 mW (0805 rating 125 mW).
D1–D8: XINGLIGHT XL-2835WWC-02 warm white 2835 (2800–3000 K), LCSC C2843875
R1–R8: UNI-ROYAL 0805W8F1000T5E 100R 1% 0805, LCSC C17408



USB-C sink: 5.1k pull-downs on CC1 and CC2 tell a USB-C charger or
C-to-C cable to turn on 5 V (each CC pin needs its own resistor).
D+/D−/SBU are not used. USB-A-to-C cables supply 5 V regardless.

Assembled by JLCPCB (economic PCBA, top side)
8x warm white 2835 LED light bar, 100 x 15 mm, USB-C power input
Banyule Nillumbik Tech School

Sheet: /
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Title: USB LED Bar

Size: A4	Date: 2026-10-15	**Rev: 2.0**
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